Khaled Masoumifard

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Stellenbosch, South Africa

Xaleed

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Skills

Programming : R, Python, Julia, C++, PyTorch, Shiny

Data Science : Statistical modeling, Machine learning

Finance : Market surveillance, Algorithmic trading strategy development & backtesting

Actuarial Science : Health insurance, Reinsurance

Optimization : Stochastic control, Finite difference methods, Majorization theory, Stochastic ordering

Languages

- Kurdish
- Persian
- English

Awards

- First place, Iran Statistics Competition, 2012
- Second place, Iran Statistics Olympiad, 2012
- National scholarship for top three statistics students, Iran, 2012
- Best Doctoral Student Award, Shahid Beheshti University, 2020
- Best Paper Award, FINACT-IRAN Conference, 2020

Quantitative Researcher
Data Scientist

Profile

Statistician with a background spanning both academic and industrial environments. Applied experience includes machine learning, actuarial science, and market surveillance in securities markets. Proficient in programming and library development using R, Python, Julia, and C++. I approach problems with rigor and curiosity, combining theoretical insight with practical application.

Experience

Postdoctoral Fellow at MuViSU, Centre for Multi-dimensional Data Visualisation, Stellenbosch University *2024 – present*

Developing machine learning libraries for compositional and spatial data, with research in statistical modeling and algorithm development.

Data Scientist at Securities and Exchange Organization of Iran *2022 – 2024*

Monitored large-scale trade transaction data (thousands per minute) to detect fraud and market manipulation. Developed time-series models and deep learning architectures (LSTM, RNN) using Python, R, and Oracle BI.

Data Scientist roles at insurance companies: Mellat Insurance Company (2019 – 2021) and Tejaratno Company (Part-time, 2023 – 2024), Tehran, Iran

Performed actuarial modeling for health insurance and reinsurance, covering rate-making, predictive modeling, and analytics. Led healthcare pricing models for short- and long-term products, co-supervised two PhD researchers, and developed an interactive Shiny dashboard later deployed via a Django framework.

Lecturer (part-time) at Department of Mathematical Sciences, Shahid Beheshti University, Tehran, Iran *2021 – 2022*

Taught and supervised students in statistics and actuarial science.

Quantitative Trading Researcher (Freelance) *2020 – 2022*

Contributed to a research-based project developing a dynamic algorithm for automated high- and low-frequency trading. Designed mathematical models, applied deep learning, and handled large-scale online trading data.

Education

Ph.D., Statistics, Shahid Beheshti University, Tehran, Iran *2021*

Thesis: Stochastic Optimization of the Dividend, Investment and Reinsurance Strategies

M.Sc., Statistics, University of Tehran, Tehran, Iran *2015*

Thesis: Stochastic Comparisons of Series and Parallel Systems with Generalized Exponential Components

B.Sc., Statistics, Razi University, Iran *2012*

References

Sugnet Gardner-Lubbe
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Stellenbosch University
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Stephan van der Westhuizen
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Selected Publications

Dirichlet-random forest for predicting compositional data
Masoumifard, K., van der Westhuizen, S., Gardner-Lubbe, S. (2026). In A. Bekker, P. Nagar, J. Ferreira, B. Erasmus, & A. Ramoelo (Eds.), *Environmental modelling with contemporary statistics: Learning, directionality, and space-time dynamics*. CRC Press. ISBN: 9781032903910. [In press]

Hazard Rate Order Between Parallel Systems With Multiple Types of Scaled Components
Masoumifard, K., Haidari, A., Torrado, N. (2025). Applied Stochastic Models in Business and Industry, 41(2), e70008.

On Bayesian credibility mean for finite mixture distributions
Jahanbani, E., Najafabadi, A.T.P., Masoumifard, K. (2023). Annals of Actuarial Science, 1–25.

Optimal dynamic reinsurance strategies in multidimensional portfolio
Masoumifard, K., Zokaei, M. (2021). Stochastic Analysis and Applications, 39(1), 1–21.

Stochastic comparisons of series and parallel systems with generalized exponential components
Balakrishnan, N., Haidari, A., Masoumifard, K. (2014). IEEE Transactions on Reliability, 64(1), 333–348.

Submitted Papers

Random forest models for compositional data
Masoumifard, K., van der Westhuizen, S., Lubbe, S. (2026). *South African Statistical Journal*. [Under review]

A machine learning framework for modelling compositional soil properties in digital soil mapping
van der Westhuizen, S., Masoumifard, K. (2026). *Mathematical Geosciences*. [Under review]

Software

DirichletRF: Dirichlet Random Forest (R package, CRAN)
Masoumifard, K., van der Westhuizen, S., Lubbe, S. (2026). Version 0.1.0. DOI: 10.32614/CRAN.package.DirichletRF

Conference Presentations

Dirichlet-random forest for predicting compositional data
Masoumifard, K., van der Westhuizen, S., Gardner-Lubbe, S. (2025). *Book of Abstracts, Data Science, Statistics & Visualisation (DSSV) Conference*, JDSSV. [Link] [International]

Random forest model for compositional data
Masoumifard, K. (2025). *66th Annual Conference of the South African Statistical Association (SASA)*. [National]